

Digital Poster Upload & Offline Display System

Professional Architecture Overview

1. System Objective

This system allows delegates to upload posters from anywhere via the internet, while ensuring that all live poster displays at the event venue operate fully offline. Posters are pre-processed and cached locally to guarantee instant display and zero dependency on internet connectivity during the event.

2. Three-Software Architecture

The solution is intentionally divided into three independent software systems. Each software has a single responsibility, making the system reliable, scalable, and easy to operate.

- 1 QR Upload System (Cloud): Handles poster submission and token generation.
- 2 Admin Dashboard (Local Server): Controls the event, syncs posters, and manages screens offline.
- 3 Display UI (Screen Laptops): Shows posters in fullscreen mode using local cached files.

3. End-to-End Workflow

- 1 Delegates upload posters via a QR-based web form hosted online.
- 2 The cloud system stores the original file and generates a unique token.
- 3 The local admin server downloads and converts posters before the event.
- 4 Converted posters are pushed and cached on each screen laptop.
- 5 Delegates enter their token on any screen to instantly display their poster.

4. Offline Reliability & Performance

All posters are stored and served locally during the event. Each screen laptop keeps its own copy of posters, ensuring that multiple screens can display the same poster simultaneously without any network congestion or delay.

5. Security & Stability

- 1 Only the upload system is exposed to the internet.
- 2 Admin and display systems operate on a closed local network.
- 3 No cloud dependency during live sessions.

6. Deployment & Operations

The system is deployed using standard laptops, a local router, and browser-based interfaces. No mobile applications or specialized hardware are required. This makes the solution portable, cost-effective, and easy to set up at different venues.

7. Summary

This architecture combines the convenience of online uploads with the reliability of offline event execution. It is designed specifically for conferences, universities, and exhibitions where stability, speed, and professional presentation are critical.